DATA ANALYTICS PROJECT REPORT

Facebook vs AdWords

A/B Testing Analysis of 2019 Ad Campaigns

Which ad platform delivers more conversions,
more clicks and better cost-effectiveness?

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Executive Summary

This report compares two digital advertising campaigns that ran every day of 2019: one on Facebook and one on Google AdWords. It asks which platform is more effective in terms of conversions, clicks and overall cost-effectiveness, so that client budgets can be allocated where they work hardest.

Key conclusion

Bottom line: Facebook is the stronger platform. It produced about twice as many conversions per day (11.74 vs 5.98) and cost roughly one-third as much per conversion ($7.48 vs $22.57). AdWords attracts more clicks, but far fewer of them turn into conversions.

Metric (2019 totals)	Facebook	AdWords
Total conversions	4,286	2,183
Total spend	$32,040	$49,266
Conversion rate (conversions / clicks)	26.7%	9.9%
Click-through rate (clicks / views)	2.02%	1.28%
Cost per click	$1.99	$2.24
Cost per conversion	$7.48	$22.57

Main findings

- **Hypothesis test:** a one-tailed Welch t-test shows Facebook converts significantly more than AdWords ($t = 32.88$, $p < 0.001$). The estimated gap is 5.76 conversions per day (95% CI 5.42 to 6.11).
- **Clicks are not equal:** clicks and conversions are strongly related on Facebook ($r = 0.87$) but only moderately on AdWords ($r = 0.45$).
- **Regression:** on Facebook, each extra click is associated with about 0.21 extra conversions ($R^2 = 76.3\%$). The same model explains only 20.1% of AdWords conversions.
- **Time trend:** Facebook conversions rose from 334 in January to 413 in December (+24%), while AdWords stayed flat.
- **Cost-effectiveness:** Facebook's monthly cost per conversion stayed between $6.88 and $8.38; AdWords stayed between $21.55 and $23.93.

Recommendation in one line

Shift budget toward Facebook, diagnose why AdWords clicks convert so poorly before funding it further, and confirm the result with a randomized budget-split test in 2020.

1. Business Problem and Objectives

As a marketing agency, the primary objective is to maximize the return on investment (ROI) of clients' advertising campaigns. Two campaigns were run in 2019, one on Facebook and one on AdWords. To allocate resources efficiently and improve client outcomes, the agency needs to know which platform yields better results in terms of clicks, conversions and overall cost-effectiveness.

Objectives

- Compare the two platforms on clicks, conversions and cost using a full year of daily data.
- Test statistically whether the difference in conversions is real rather than random variation.
- Quantify how clicks translate into conversions so that campaign goals can be set realistically.
- Understand how performance and cost change across months and days of the week.
- Turn the results into a clear budget-allocation recommendation.

2. Research Question

Primary research question

Which ad platform is more effective in terms of conversions, clicks, and overall cost-effectiveness?

Supporting questions

- How often does each platform reach high-conversion days compared with low-conversion days?
- Do more clicks really lead to more conversions, and is that equally true on both platforms?
- Is the Facebook conversion advantage statistically significant?
- How many Facebook conversions can be expected for a given number of clicks?
- How do conversions and cost per conversion move over the year and across weekdays?

3. Data Description

The dataset (marketing_campaign.csv) holds one row for every day of 2019, from 1 January to 31 December, giving 365 rows and 17 columns. Each row contains the daily performance of both campaigns side by side. Both campaigns are split into monthly campaign codes (for example FB_Jan19 and AW_Jan19).

3.1 Field dictionary

Field	Description
Date	Calendar day of the observation (1 Jan to 31 Dec 2019)
Facebook / AdWords Ad Campaign	Monthly campaign code (12 per platform)
Ad Views	Number of times the ad was shown
Ad Clicks	Number of clicks received
Ad Conversions	Number of conversions resulting from the ad
Cost per Ad	Amount spent on the platform that day (in US dollars)
Click-Through Rate (CTR)	Clicks divided by views
Conversion Rate	Conversions divided by clicks
Cost per Click (CPC)	Ad cost divided by clicks

3.2 Data preparation and quality

- The Date column was converted to a datetime type.
- Cost, CTR, conversion rate and cost-per-click columns arrived as text with $ and % symbols; these were stripped and converted to numbers.
- There are no missing values and no duplicate rows (365 complete rows).
- The supplied cost-per-click column differs slightly from cost divided by clicks (on average by about 0.6% per day), most likely because of rounding in the source. For this reason, all cost metrics in this report are recomputed from total cost and total clicks or conversions.

3.3 Descriptive statistics

Daily metric	Platform	Mean	Std dev	Min	Max
Ad views	Facebook	2,179.7	618.1	1,050	3,320
	AdWords	4,717.2	561.1	3,714	5,760
Ad clicks	Facebook	44.0	12.1	15	73
	AdWords	60.4	14.4	31	89
Ad conversions	Facebook	11.7	2.9	5	19
	AdWords	6.0	1.6	3	9
Daily spend ($)	Facebook	87.8	28.3	34	144
	AdWords	135.0	32.3	73	197

Table 1. Daily descriptive statistics for each platform (n = 365 days).

AdWords reaches roughly twice the audience (about 4,717 views a day vs 2,180) and attracts more clicks per day (60.4 vs 44.0), but it also spends more ($135 vs $88 a day) and records only about half as many conversions.

4. Methodology

The analysis follows five steps, each building on the previous one.

Step	What was done	Tools
1. Prepare	Loaded 365 daily rows, parsed dates, converted $ and % text fields into numbers	pandas
2. Explore	Plotted distributions of clicks and conversions, grouped days into conversion buckets, and measured the click-to-conversion correlation	pandas, seaborn, matplotlib
3. Test	Welch two-sample t-test (unequal variances), one-tailed, significance level $\alpha = 0.05$	SciPy
4. Model	Simple linear regression of conversions on clicks, evaluated with R^2 and mean squared error (MSE)	scikit-learn
5. Trend	Aggregated by month and weekday, computed cost per conversion, ran an Engle–Granger cointegration test between cost and conversions	statsmodels

Key metric definitions

Metric	Formula
Click-through rate (CTR)	Clicks ÷ Views
Conversion rate	Conversions ÷ Clicks
Cost per click (CPC)	Total cost ÷ Total clicks
Cost per conversion	Total cost ÷ Total conversions (the notebook labels this "CPC")

Rates in this report are calculated from 2019 totals (for example total conversions divided by total clicks) rather than by averaging daily percentages, which avoids giving low-traffic days too much weight.

5. Analysis and Findings

5.1 Distribution of clicks and conversions

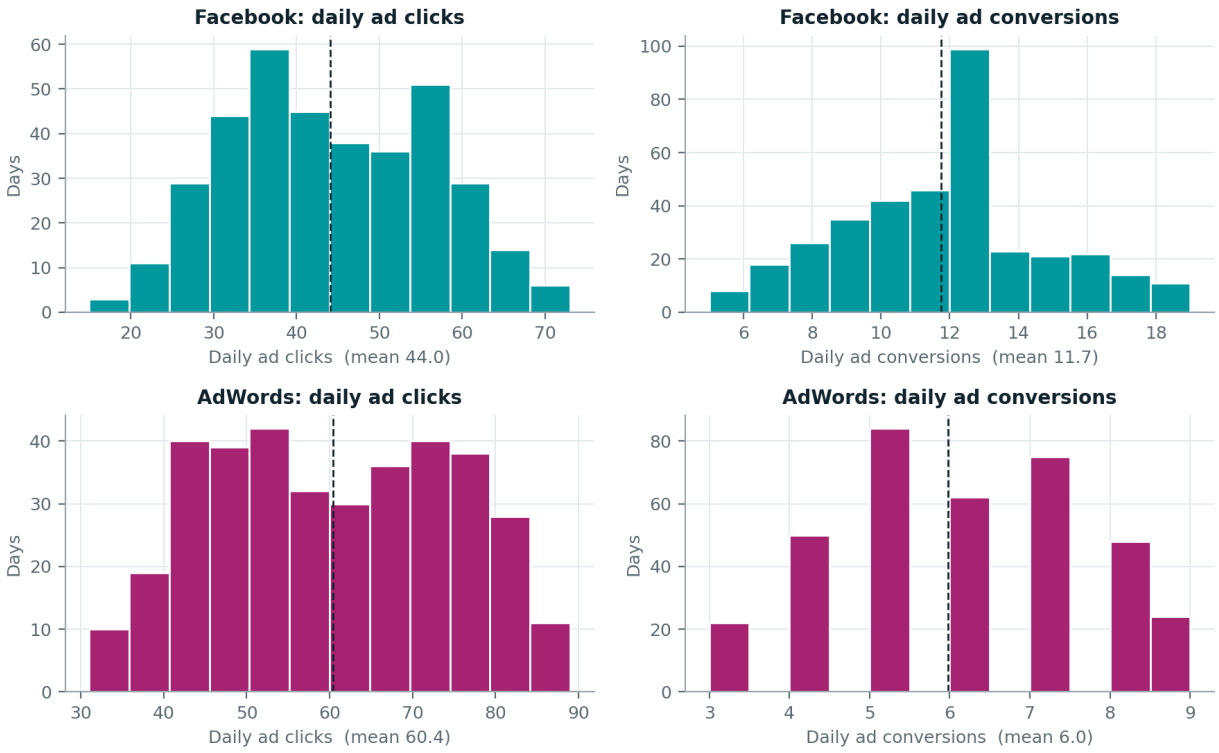


Figure 1. Daily clicks and conversions are roughly symmetric on both platforms. Dashed lines mark the means.

All four distributions are broadly symmetric with no extreme outliers, so the mean is a fair summary of a typical day. The two platforms, however, sit at very different levels: Facebook averages 11.7 conversions a day and AdWords 6.0.

5.2 How often do high-conversion days occur?

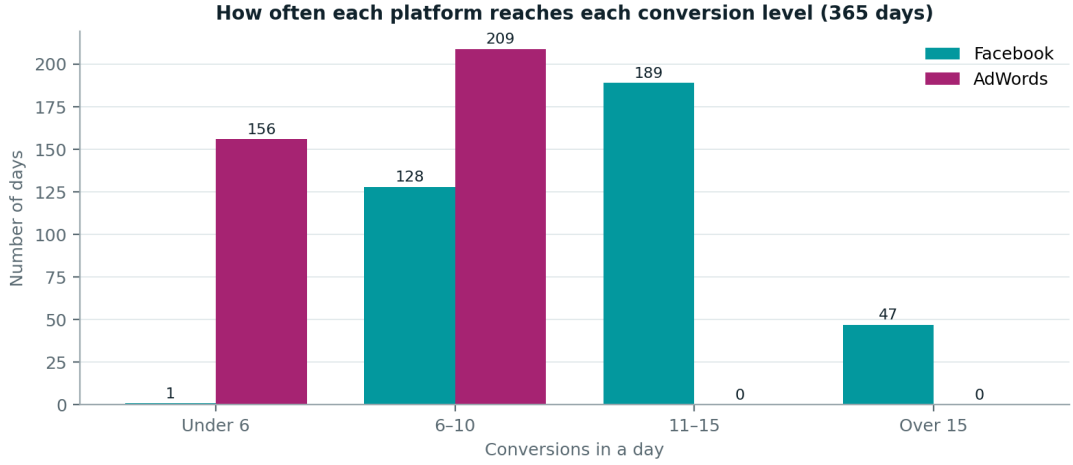


Figure 2. Number of days falling into each daily-conversion bucket.

Facebook reached 11 or more conversions on 236 of 365 days (65%), including 47 days above 15. AdWords never reached 11: it recorded under 6 conversions on 156 days (43%) and 6–10 on the remaining 209. Facebook fell below 6 on a single day. This is a structural gap, not a handful of lucky days.

5.3 Platform scorecard

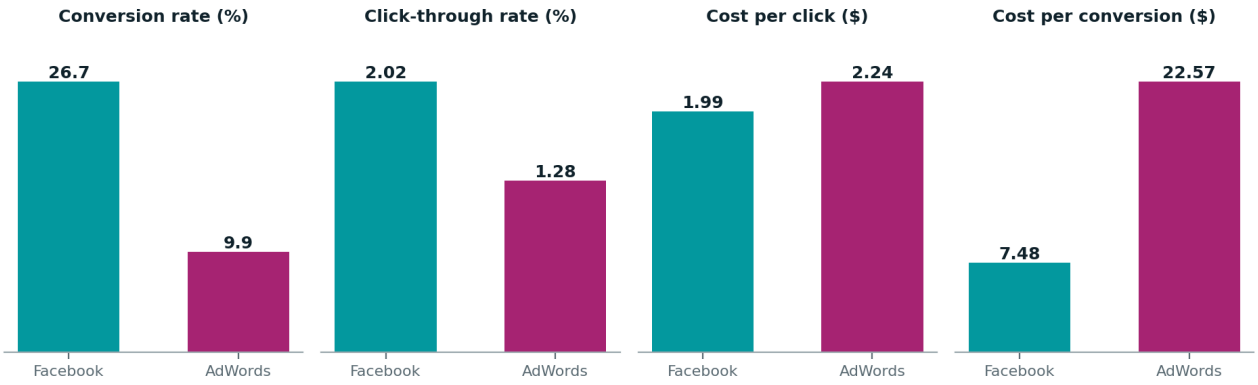


Figure 3. Efficiency metrics computed from 2019 totals.

Facebook leads on every efficiency measure. Its click-through rate is 2.02% against 1.28%, so its ads earn a click more often. Once a visitor has clicked, Facebook converts 26.7% of them against 9.9% on AdWords, about 2.7 times better. A click also costs less ($1.99 vs $2.24). Combined, these give Facebook a cost per conversion of $7.48, about 3.0 times cheaper than AdWords at $22.57.

5.4 Do more clicks lead to more conversions?

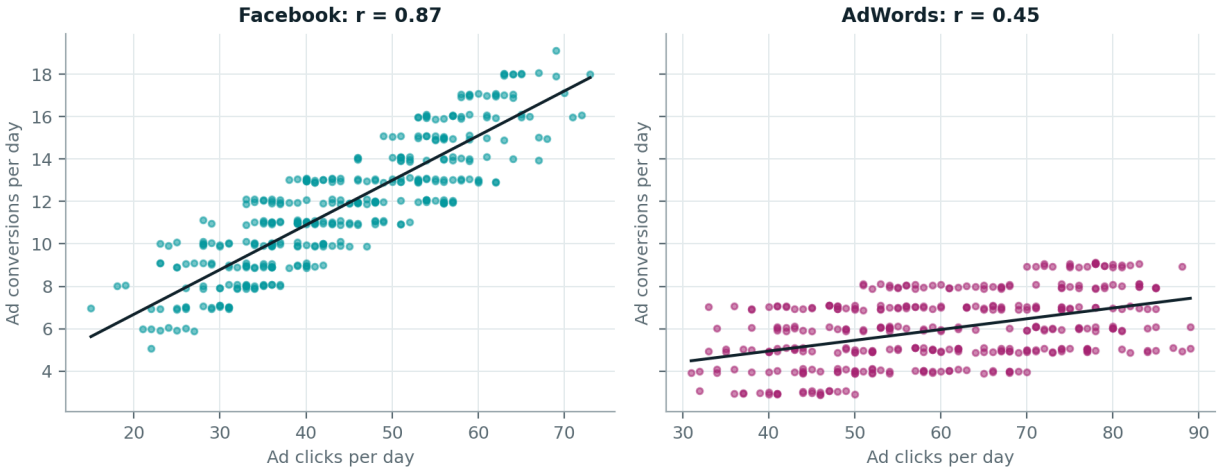


Figure 4. Clicks vs conversions per day (points jittered slightly for visibility), with a straight-line trend.

On Facebook the relationship is strong (r = 0.87): busier click days reliably produce more conversions, so extra traffic is worth buying. On AdWords it is only moderate (r = 0.45): many additional clicks do not translate into additional conversions, which suggests that other factors, such as landing-page quality, keyword intent or audience fit, limit its effectiveness.

6. Hypothesis Testing

The question of whether Facebook converts better is tested formally so that the conclusion does not depend on chance variation between days.

	Statement
Null hypothesis (H0)	μ Facebook ≤ μ AdWords: there is no difference in conversions, or AdWords converts at least as much
Alternative hypothesis (H1)	μ Facebook > μ AdWords: Facebook produces more conversions
Test	Welch two-sample t-test (does not assume equal variances), one-tailed
Significance level	α = 0.05

Results

Statistic	Value
Mean conversions per day, Facebook	11.74
Mean conversions per day, AdWords	5.98
Difference in means	5.76 (95% CI 5.42 to 6.11)
t statistic	32.88
p-value	9.3e-134
Effect size (Cohen's d)	2.43
Decision	Reject H0

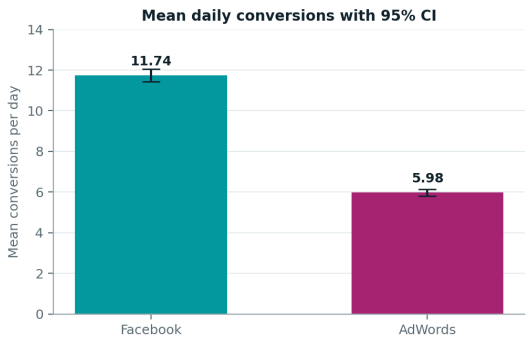


Figure 5. Mean daily conversions; error bars show the 95% confidence interval and are barely visible because the intervals are narrow.

The p-value is far below 0.05, so the null hypothesis is rejected. The result is not only statistically significant but practically large: Cohen's d of 2.4 is well above the conventional threshold of 0.8 for a large effect, and the gap of about 5.8 conversions a day is roughly double AdWords' entire daily average.

Important nuance

A supplementary test on clicks points the other way: AdWords receives significantly more clicks per day (60.4 vs 44.0, $p < 0.001$). AdWords wins on traffic volume, while Facebook wins on what that traffic achieves.

7. Regression Analysis

If Facebook is chosen, how many conversions can be expected for a given number of clicks? A simple linear regression with clicks as the independent variable and conversions as the dependent variable answers this.

Item	Facebook	AdWords (for comparison)
R^2 (variance explained)	76.35%	20.1%
Mean squared error	2.02	2.11
Slope (conversions per extra click)	0.211	0.051
Intercept	2.47	2.92

Fitted Facebook model

Conversions ≈ 2.47 + 0.21 × Clicks. As a rule of thumb, every 5 extra clicks add about 1 conversion.

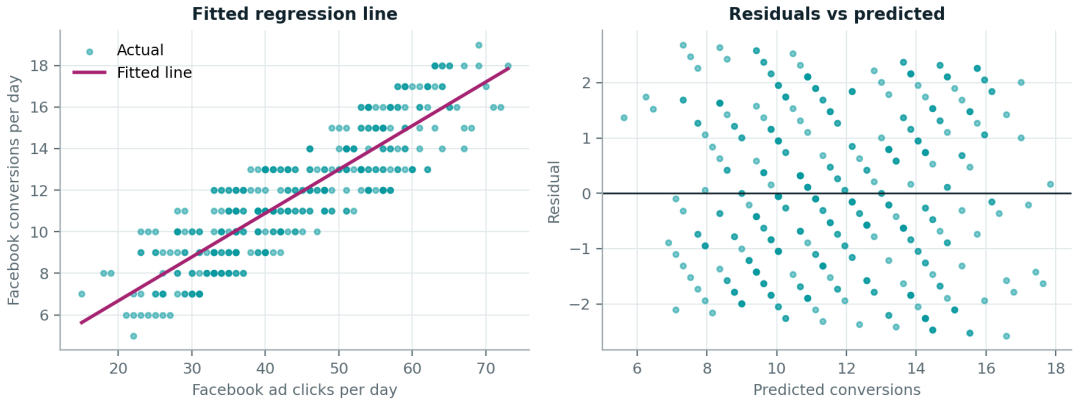


Figure 6. Left: fitted line over the data. Right: residuals scatter around zero without an obvious pattern.

Predictions

Facebook clicks in a day	Expected conversions
50 clicks	13.0
80 clicks	19.3

The model explains 76% of the day-to-day variation in Facebook conversions, which is good predictive power for a single-variable model. The residual plot shows errors spread evenly around zero, so a straight line is a reasonable description. It also helps planning: a click target becomes a conversion target and, with cost per click, a budget. The model was fitted and scored on the same 365 days, so its accuracy on future data may be somewhat lower. On AdWords the same model explains only 20% of variation, confirming that clicks are a much weaker guide to AdWords results.

8. Time Series Analysis

8.1 Long-run trend

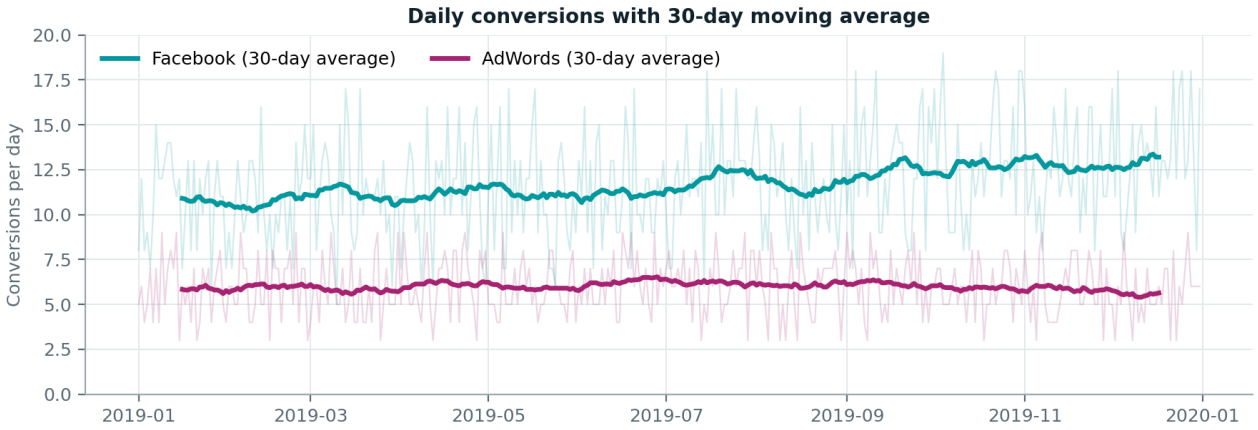


Figure 7. Daily conversions (faint) and 30-day moving average (bold).

Facebook conversions trend upward through the year: the fitted daily trend is +0.0075 conversions per day each day ($p < 0.001$), equal to roughly +2.7 conversions a day between the start and end of the year. AdWords shows no trend at all ($p = 0.88$); its performance in December is the same as in January.

8.2 Monthly conversions and cost per conversion

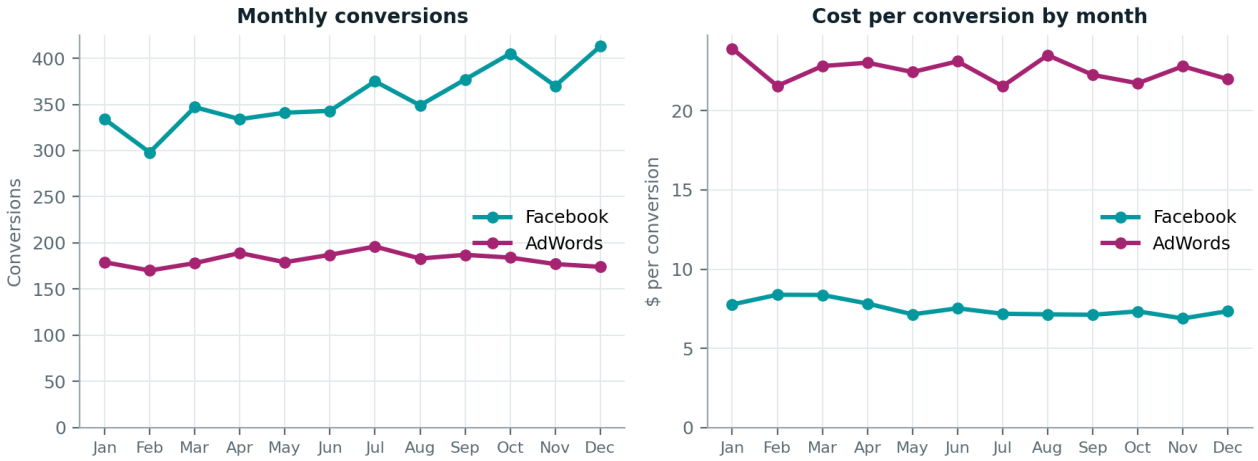


Figure 8. Left: total conversions by month. Right: monthly cost per conversion.

Facebook monthly conversions climbed from 334 in January to 413 in December, with the strongest months being October (405) and December (413). Dips appear in February (298, partly because the month has only 28 days), April, August and November. Facebook's cost per conversion ranged from $6.88 (November) to $8.38 (February), staying in a narrow band. AdWords stayed between $21.55 and $23.93, always roughly three times higher.

8.3 Day-of-week pattern

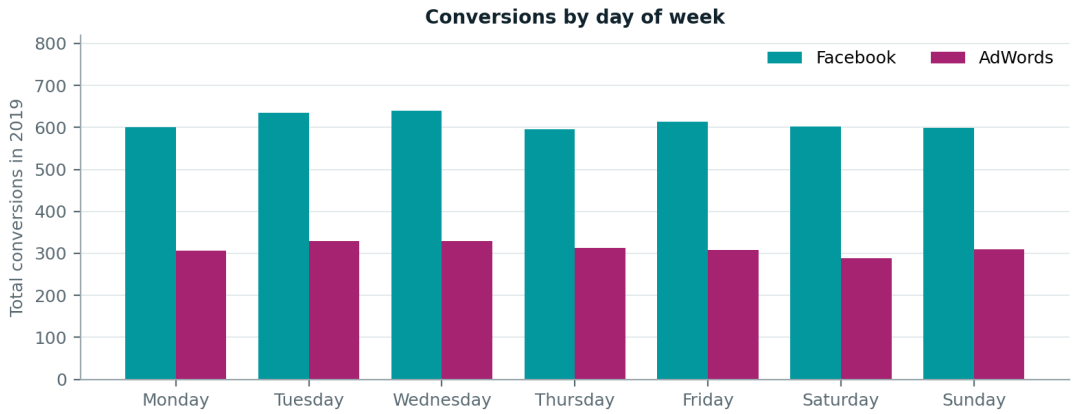


Figure 9. Total conversions in 2019 by day of week.

Conversions are spread evenly across the week. On Facebook, Tuesday (635) and Wednesday (639) are the highest and Thursday (596) the lowest, but every day lies within about 4% of the weekly average. Weekday timing is therefore a minor lever compared with platform choice.

8.4 Is there a long-term relationship between spend and conversions?

An Engle–Granger cointegration test between daily cost and Facebook conversions returned a test score of −14.76 and a p-value of about 2e-26, rejecting the null hypothesis of no cointegration.

Interpretation caveat

The test result should be read cautiously. Both series fluctuate around stable levels, which makes this test reject easily, and the direct correlation between daily spend and daily conversions is weak (r = -0.13). The data therefore do not show that raising spend on a given day raises conversions. Budget decisions should rest on cost per conversion and click-to-conversion efficiency rather than on this test alone.

9. Recommendations

1. Shift budget toward Facebook

Facebook delivers about twice the daily conversions (11.7 vs 6.0) at one-third of the cost per conversion ($7.48 vs $22.57). As an illustration, if $10,000 of spend earned conversions at the 2019 average rates, it would yield about 1,338 conversions on Facebook against about 443 on AdWords. Real returns will diminish as spend scales, so move budget in stages and monitor results.

2. Plan Facebook volume with click targets

Use the regression (about 0.21 conversions per click) to convert conversion goals into click and budget targets: for example, 80 clicks a day should yield around 19 conversions. Revisit the model regularly as the relationship may change.

3. Diagnose AdWords before funding it

AdWords brings volume (22,040 clicks) but only 9.9% convert. Audit keywords and search intent, ad-to-landing-page match, landing-page speed and offer, and audience targeting. Re-test with a small budget once changes are made.

4. Time the spend

Facebook momentum built through the year and peaked in Q4, and conversions are steady across weekdays. Track monthly cost per conversion and lean into historically low-cost months such as November, while avoiding assumptions that spending more on a given day will lift that day's conversions.

5. Validate with a controlled test

Run a randomized budget-split test in 2020, with comparable audiences and the same offer on both platforms, to confirm the Facebook advantage and to measure how returns change as spend increases.

10. Limitations and Further Work

- **Not a randomized experiment.** The two platforms reach different audiences and were not randomly assigned, so part of the gap may reflect audience differences rather than platform quality alone.
- **One year, one campaign pair.** Results are for 2019 and may not hold in other periods, markets or creative formats.
- **Conversions are not revenue.** The data has no order value or lifetime value. A conversion on one platform may be worth more than on the other, so ROI in dollars cannot be confirmed from this dataset.
- **Constant returns assumed.** Cost per conversion at 2019 spend levels may not hold if spend is scaled up sharply.
- **Simple models.** The regression uses one predictor and was not tested on held-out data. Future work could add spend, views and calendar effects, and validate on a holdout period.
- **Data rounding.** Spend values are whole dollars and the supplied cost-per-click column does not exactly match cost divided by clicks.

Appendix: Notes on the source notebook

The following refinements were made relative to the original notebook so that the report is accurate and complete.

Item	Treatment in this report
Weekday labels	The notebook labels the first bar "Sunday", but pandas weekday 0 is Monday. This report uses the correct labels, so the peak days are Tuesday and Wednesday.
"10–15" bucket	The notebook's bucket covers 11 to 15 conversions (values 6–10 are in the previous bucket); it is labelled 11–15 here.
Cost per conversion	The notebook labels this "CPC" and computes it only for Facebook. Here it is called cost per conversion and computed for both platforms.
Cointegration	Result reported as in the notebook, with an added caveat about weak direct correlation between spend and conversions.
Additional checks	Descriptive statistics for both platforms, confidence interval and effect size for the t-test, AdWords regression, and trend tests were added from the same dataset.